INCH POUND

MIL-C-915/83A
29 September 1989
SUPERSEDING
MIL-C-915/83
18 June 1985

MILITARY SPECIFICATION SHEET

# CABLE, POWER ELECTRICAL, SEVEN TWISTED PAIRS NO. 16 GAUGE FLUOROPOLYMER INSULATION, FOR SUBMARINE OUTBOARD USE ONLY (NOT FOR INBOARD USE), TYPE 7PR-16

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specifiction listed in that issue of the department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-915.

REQUIREMENTS:

Qualification required.

## Construction

First - Copper conductor, tin coated, AWG size 16, class B stranding.
Second - Insulation of ethylene tetrafluoroethylene (ETFE). Nominal wall thickness 0.010 inch. Maximum overall diameter 0.082 inch.
Third - Two insulated conductors, one black and one white, twisted together with a nominal lay of 2-1/2 inches to form each pair.
Fourth - Marker braid on each pair, standard identification code applied by method 4.
Fifth - Seven pairs cabled with a lay not greater than 24 times the pitch diameter of the layer.
Sixth - Rubber filler in order to form a firm, round assembly. Filler shall be capable of removal from the assembly without damaging the insulation on conductors.
Seventh - A binder tape of polyester, applied helically with overlap.

AMSC N/A FSC 6145
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REQUIREMENTS - Continued

Eighth - Cable jacket of black polyether polyurethane. The cable jacket material shall be B.F. Goodrich Co. ESTANE 58300 or 58863; or Mobay Chemical Co. TEXIN 985A. Nominal wall thickness 0.100 inch. Overall cable diameter to be 0.630 to 0.650 inch.

EXAMINATION AND TESTS:

| | Requirements |
|---|---|
| Basic electrical: | |
| Conductor resistance - at 20°C, ohms/1000 feet, maximum | 4.24 |
| Voltage withstand - volts, root mean square, minimum | |
| Conductor to conductor | 2000 |
| Conductor to water | 2000 |
| Insulation dielectric (on 100 percent of insulated conductor) - test in accordance with MIL-W-81381, using a test voltage of 8.0 kV (peak), minimum. | |
| Insulation resistance - megohms/1000 feet, minimum | 5000 |
| Group A: | |
| Visual and dimensional | No failure |
| Group B: | |
| Cold bending, cable - at minus 40 ± 2°C over a 4-inch diameter mandrel | No damage |
| Physicals (unaged) | |
| Insulation | |
| Tensile strength - lbf/in², minimum | 5000 |
| Elongation - percent, minimum | 100 |
| Cable jacket | |
| Tensile strength - lbf/in², minimum | 4000 |
| Elongation - percent, minimum | 500 |
| Group C: | |
| Physicals (aged) | |
| Insulation | |
| Air oven (168 hours at 180°C) | |
| Tensile strength - percent of unaged, minimum | 80 |
| Elongation - percent of unaged, minimum. | 90 |
| Cable jacket | |
| Air oven (168 hours at 80°C) | |
| This test shall be in accordance with MIL-C-915 | No failure |
| Hot oil immersion | |
| Tensile strength, percent of unaged, minimum | 60 |
| Elongation, percent of unaged, minimum | 80 |



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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, plus all of groups A, B and C.

UNIT ORDERING LENGTH: 500 feet (minimum).

NOTE: Not for Air Force use.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1127-20)

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG



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